



ENGINEERING MATHEMATICS - II

Dr. G. K. Jagatheswari

Department of Science and Humanities

ENGINEERING MATHEMATICS - II

UNIT 5 : Fourier series

Session : 2

Sub Topic : Fourier Series of Arbitrary Period and Fourier Series of Even and Odd Functions

Department of Science and Humanities

Recall!!!

We know that, for a periodic function $f(x)$ of period 2π , defined over $(0, 2\pi)$, the Fourier series expansion is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad \dots \dots \dots (1)$$

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx \quad \text{for } n = 1, 2, 3 \dots$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx dx \quad \text{for } n = 1, 2, 3 \dots$$

Recall!!

We know that, for a periodic function $f(x)$ of period 2π , defined over $(-\pi, \pi)$, the Fourier series expansion is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad \dots \dots \dots (1)$$

where $a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx.$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx, \quad n = 1, 2, 3 \dots$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx, \quad n = 1, 2, 3 \dots$$

Fourier series of functions having arbitrary period $2l$

Fourier series expansion of $f(x)$ on the interval $(-l, l)$

By taking the substitution $y = \frac{\pi x}{l}$ or $x = \frac{ly}{\pi}$

$$\therefore dy = \frac{\pi dx}{l}; \text{ Also, } y = -\pi, x = -l \\ y = \pi, x = l$$

$$F(y) = f\left(\frac{ly}{\pi}\right) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos ny + b_n \sin ny) \text{ where}$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} F(y) dy = \frac{1}{\pi} \int_{-\pi}^{\pi} f\left(\frac{ly}{\pi}\right) dy.$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} F(y) \cos ny dy = \frac{1}{\pi} \int_{-\pi}^{\pi} f\left(\frac{ly}{\pi}\right) \cos ny dy.$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} F(y) \sin ny dy = \frac{1}{\pi} \int_{-\pi}^{\pi} f\left(\frac{ly}{\pi}\right) \sin ny dy.$$

Fourier series of functions having arbitrary period $2l$

Now, substituting for y in terms of x , we have

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right)$$

where

$$a_0 = \frac{1}{l} \int_{-l}^l f(x) dx.$$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx$$

$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx$$

Fourier series of functions having arbitrary period $2l$

The Fourier series expansion of a function $f(x)$ defined over an interval $(-l, l)$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right)$$

where

$$a_0 = \frac{1}{l} \int_{-l}^l f(x) dx.$$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx$$

$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx$$

Fourier series of functions having arbitrary period $2l$

The Fourier series expansion of a function $f(x)$ defined over an interval $(0, 2l)$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right)$$

where

$$a_0 = \frac{1}{l} \int_0^{2l} f(x) dx.$$

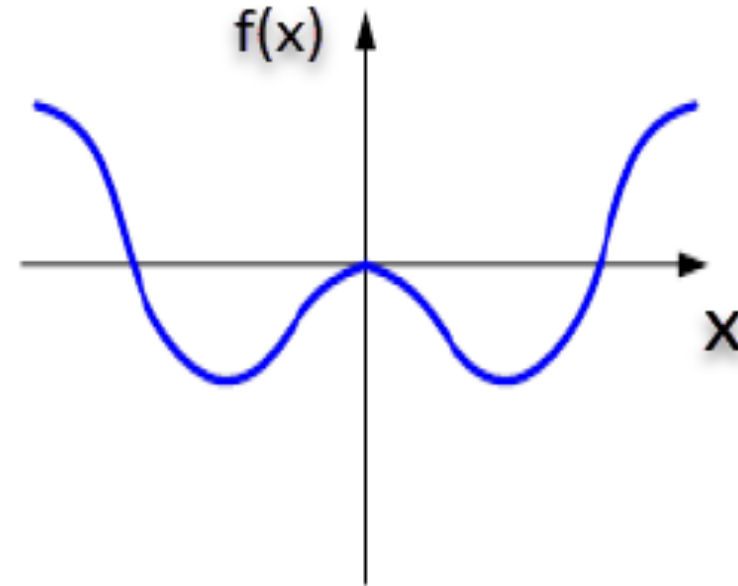
$$a_n = \frac{1}{l} \int_0^{2l} f(x) \cos \frac{n\pi x}{l} dx$$

$$b_n = \frac{1}{l} \int_0^{2l} f(x) \sin \frac{n\pi x}{l} dx$$

Even Function

A function $f(x)$ is said to be an **even function** if $f(-x) = f(x)$.

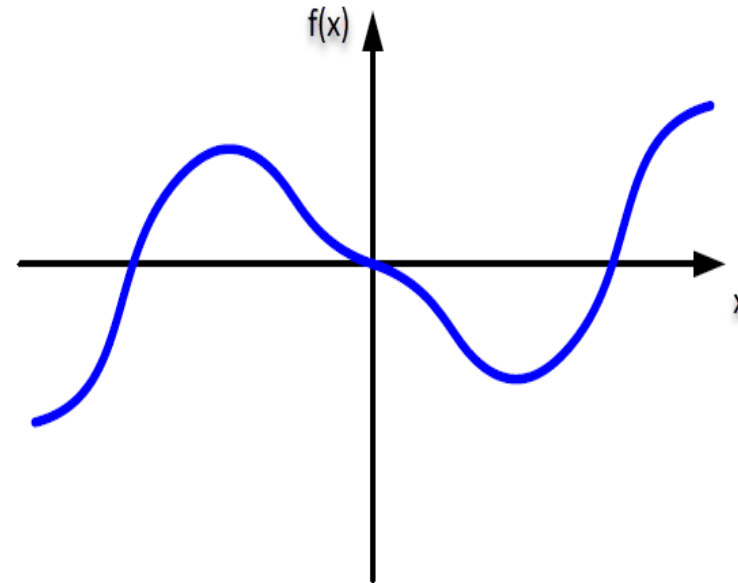
Graphically an even function is **symmetrical about the y-axis**



Odd Function

A function $f(x)$ is said to be an **odd function** if $f(-x) = -f(x)$.

Graphically an odd function is **symmetrical about the origin**.

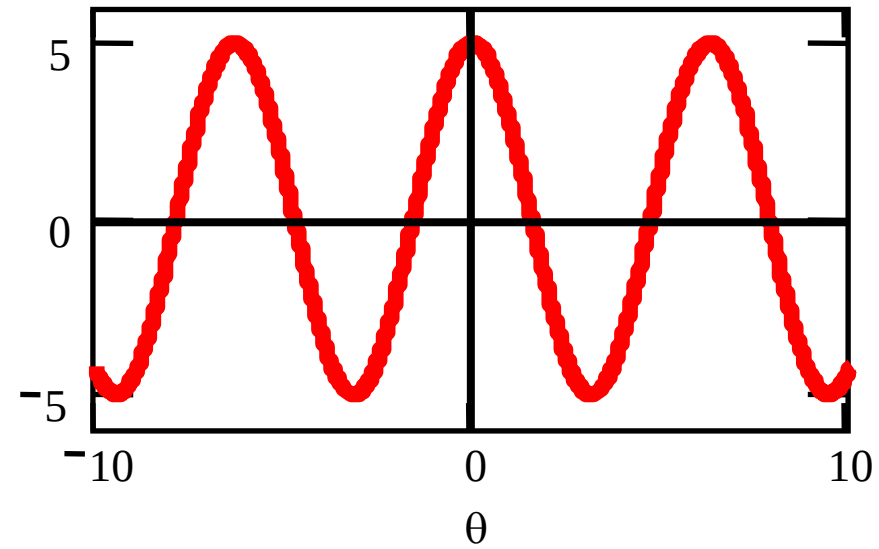


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Fourier Series of Even and Odd Functions

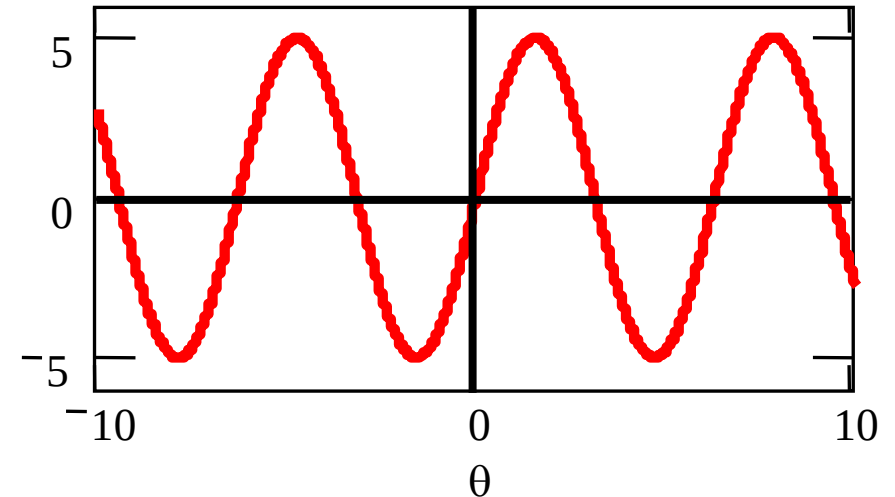
Points to remember

1. Even functions can be represented by cosine waves because, cosine waves are even functions.
2. A sum of even functions is another even function.
3. Product of two even function is even.



Points to remember

1. Odd functions can be represented by sine waves because sine waves are odd functions.
2. A sum of odd functions is another odd function.
3. Product of an odd function and even function is odd.
4. Product of two odd functions is even



Characteristics of Even and Odd Functions

	Even Function	Odd Function
1	A function $f(x)$ is said to be an even function if $f(-x) = f(x)$.	A function $f(x)$ is said to be an odd function if $f(-x) = -f(x)$.
2	Graph of an even function is symmetrical about the y-axis	Graph of an odd function is symmetrical about the x-axis
3	For an even function $\int_{-c}^c f(x)dx = 2 \int_0^c f(x)dx$	For an odd function $\int_{-c}^c f(x)dx = 0$
4	Product of two even functions is even . Product of two odd functions is even .	Product of an even and an odd function is odd .

Fourier Series of an even function in $(-\pi, \pi)$

Case 1:

Suppose $f(x)$ is an **even function** in $(-\pi, \pi)$, then the Fourier series expansion contains only cosine terms and is known as **Fourier cosine series** given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx \dots \dots \dots (2)$$

where

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{\pi} f(x) dx.$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx \, dx, n = 1, 2, 3 \dots$$

Fourier Series of an odd function in $(-\pi, \pi)$

Case 2:

Suppose $f(x)$ is an **odd function** in $(-\pi, \pi)$, the Fourier series expansion contains only sine terms and is known as **Fourier sine series** given by

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx \dots \dots \dots (3)$$

where

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx, n = 1, 2, 3 \dots$$

Fourier Series of an even function defined in $(-l, l)$

Case 1:

Fourier series of an **even function** $f(x)$ in $(-l, l)$ contains only cosine terms and is known as **Fourier cosine series** given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} \right)$$

$$\text{where } a_0 = \frac{1}{l} \int_{-l}^l f(x) dx = \frac{2}{l} \int_0^l f(x) dx$$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} dx, \quad n = 1, 2, 3, \dots$$

Fourier Series of an odd function defined in $(-l, l)$

Case 2:

Fourier series of an **odd function** $f(x)$ in $(-l, l)$ contains only sine terms and is known as **Fourier sine series** given by

$$f(x) = \sum_{n=1}^{\infty} \left(b_n \sin \frac{n\pi x}{l} \right)$$

where

$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} dx, \quad n = 1, 2, 3, \dots$$



THANK YOU

Dr. G K Jagatheswari

Department of Science & Humanities

jagatheswarigk@pes.edu